

✓ ZoyaSofts Internship Training

Week 1 — AI & Python Foundations

Day 1 — Introduction to AI

What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of creating computer systems that can perform tasks which normally require human intelligence, such as understanding language, making decisions, solving problems, and recognizing patterns.

Examples of AI

- Voice assistants
- YouTube recommendations
- Gmail spam filters
- Chatbots
- Face recognition

Why is AI important?

AI helps businesses automate tasks, improve decision-making, provide better user experiences, and solve real-world problems efficiently.

AI vs Machine Learning

Artificial Intelligence (AI)

Artificial Intelligence is the broader field of creating systems that can perform tasks that require intelligent behavior.

Machine Learning (ML)

Machine Learning is a branch of Artificial Intelligence where systems learn patterns from data and use those patterns to make predictions or decisions.

Relationship

Artificial Intelligence is the larger field, and Machine Learning is a part of AI.

AI ↓ Machine Learning

Simple Example

A spam filter that learns from previous emails can use Machine Learning to identify whether a new email is likely to be spam.

Key Difference

- AI is the broader concept.
- ML is one approach used to build AI systems.
- ML uses data to learn patterns rather than requiring every decision to be manually programmed.

Deep Learning and Generative AI

Deep Learning

Deep Learning is a branch of Machine Learning that uses neural networks with multiple layers to learn patterns from data.

Deep Learning is useful for tasks involving complex data such as images, audio, and language.

Generative AI

Generative AI refers to AI systems that can generate new content.

It can generate:

- Text
- Images
- Audio
- Code

Example

A chatbot that generates a product description from a user's request is an example of a Generative AI task.

Relationship

AI ↓ Machine Learning ↓ Deep Learning ↓ Generative AI

Generative AI is associated with the Deep Learning branch in the framework used in this internship training.

Key Difference

- Machine Learning learns patterns from data.
- Deep Learning uses multi-layer neural networks.
- Generative AI produces new content.

Narrow AI vs General AI

Narrow AI

Narrow AI is designed to perform a specific task or a limited set of tasks.

Examples include:

- Spam detection
- Recommendation systems
- Voice assistants
- Image recognition

A Narrow AI system can be very effective at its particular task, but it is not designed to have broad, general intelligence.

General AI

General AI refers to the idea of an AI system with broad, general-purpose intelligence rather than being limited to a specific task.

Key Difference

- Narrow AI → Designed for specific or limited tasks.
- General AI → Designed for broad, general intelligence.

Examples

A recommendation system that suggests videos is an example of Narrow AI because it is designed for a specific purpose.

A hypothetical system capable of handling a wide range of intellectual tasks would be closer to the idea of General AI.

Real-World AI Applications

Artificial Intelligence is used in many real-world products and services.

Examples

1. Recommendation Systems

- Recommend videos, products, movies, or other content based on user behavior and preferences.
- 2. Spam Filters
 - Identify unwanted or spam emails.
- 3. Voice Assistants
 - Understand spoken requests and respond to users.
- 4. Chatbots
 - Interact with users and provide responses using natural language.
- 5. Face Recognition
 - Analyze images and identify or verify faces.

Why AI is Used

AI can help businesses automate tasks, support decision-making, improve user experiences, and solve real-world problems.

AI Development Lifecycle

An AI project generally follows these stages:

1. Problem Definition

Identify and clearly define the problem that the AI system needs to solve.

2. Data

Collect and prepare the data required to solve the problem.

3. Model

Select or develop an appropriate AI or Machine Learning model for the problem.

4. Evaluation

Evaluate how well the system performs and check whether it meets the required goals.

5. Deployment

Put the working AI system into a real application so that users can use it.

Lifecycle

Problem Definition ↓ Data ↓ Model ↓ Evaluation ↓ Deployment

✓ Day 1 — Intern Practice Task

AI Product Analysis

1. YouTube Recommendation System

- Type: Narrow AI
- Technology: Machine Learning / Deep Learning
- Reason: It is designed for the specific task of recommending videos based on user interests and behavior.

2. Gmail Spam Filter

- Type: Narrow AI
- Technology: Classic Machine Learning
- Reason: It performs the specific task of identifying and filtering unwanted or spam emails.

3. ChatGPT

- Type: Narrow AI
- Technology: Generative AI / Deep Learning
- Reason: It generates new content such as text and code in response to user prompts.

Conclusion

The three selected products are examples of Narrow AI because they are designed for specific capabilities rather than broad general intelligence.

Start coding or generate with AI.

